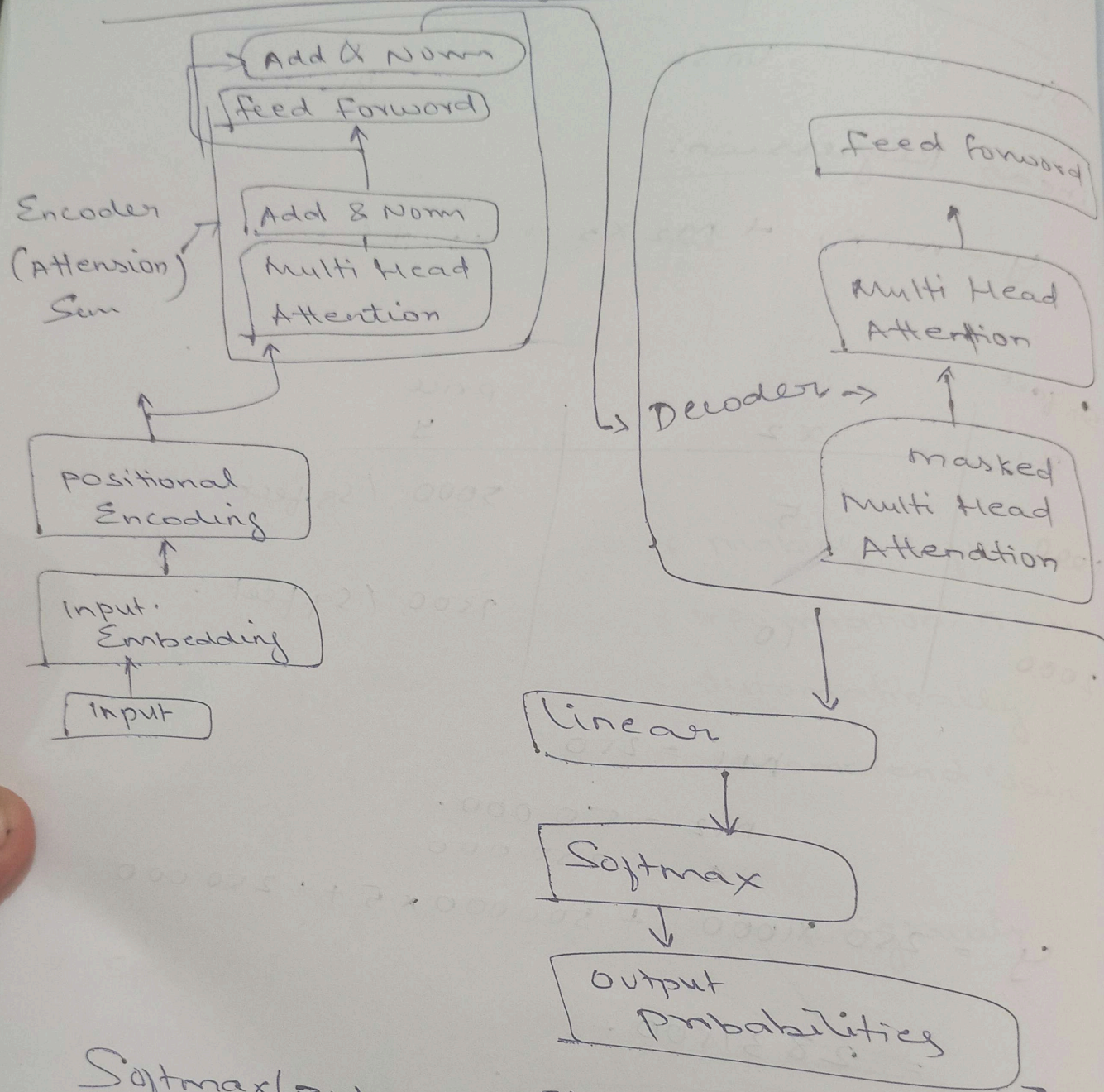


Transformer Diagram:



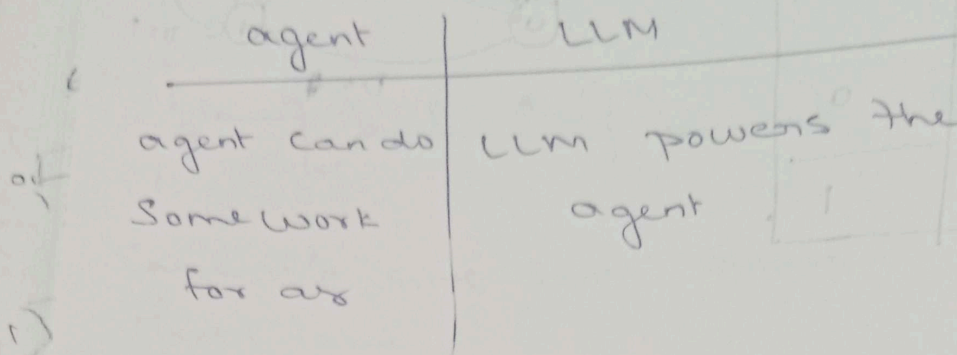
$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}}$$

The Softmax func Converts a set of numbers into probabilities where all probabilities add up to 1.

Day 1

EEC Course:

1. Intro:



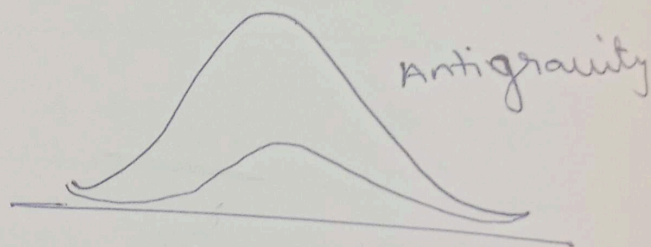
2 Software Intelligence:

Software Intelligence means making Software Smart enough to understand information, make decisions, and perform tasks automatically.

ex: Instagram intelligence is recommending using the meta data.

* feasibility

* Space.



1. What makes a System intelligent?

2. Can your PY data Survive.

3. What does a model actually learn?

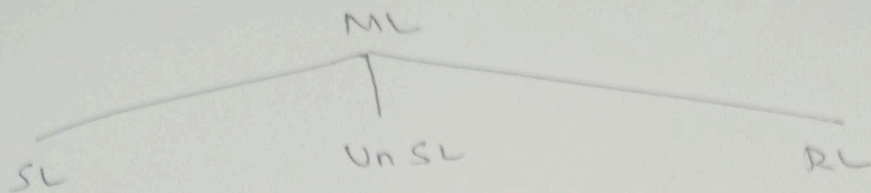
4. Why Can 95%?

json → Java Script Object Notation

multiple

keys

values data.



Linear Regression:

$$y = m_1 x_1 + m_2 x_2 + \dots + c$$

Sq. feet x_1	x_2	price y
1000	5	5000 / Sq. feet
2000	10	7500 / Sq. feet.

$$m_1 = 250$$

$$m_2 = 50000.$$

$$c = 250000$$

$$y = 250 \times 1000 + 50000 \times 5 + 250000$$

$$= 283750.$$

Training and Interface.



Understand and learn.

Some of the patterns.

Tokens.

matrix

Overfitting and Underfitting

hallucination

- Random Forest ... [10 20 30 40] | [1, 2, 3, 100]

RF [10, 3]

↓
O.T.

RF [10, 1]

↓
O.T.

RF [10, 2]

↓
O.T.

- Linear Regression

- Decision Trees.

- Catboost

- Random Forest

- SVM → eqn of hyperplane : $w_1 x_1 + w_2 x_2 + b = 0$

- KNN

- XG Boost.

SVM:

data :



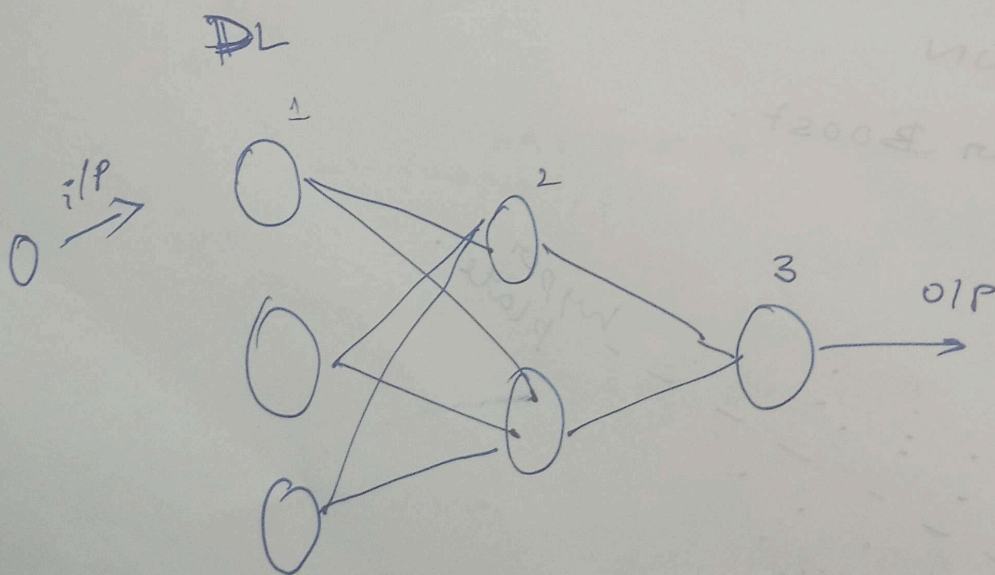
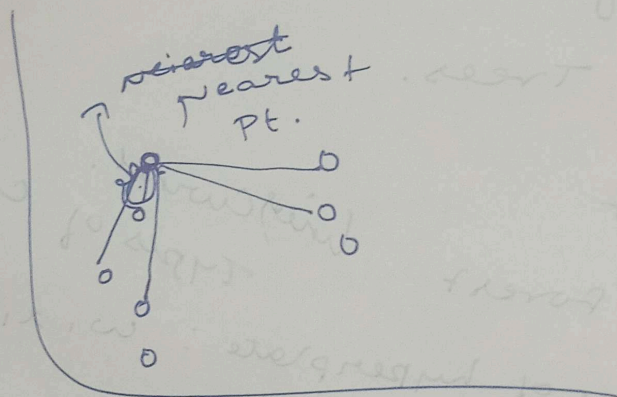
hyperplane.

$$\sum_{i=1}^n \frac{1}{n} | \text{Actual} - \text{prediction} | = \text{loss}$$

mean error

KNN - k nearest Neighbours:

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad k = 5$$



Rmse
MSE
MAE
R².

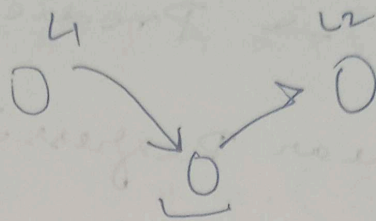
- sigmoid - $[0, 1] \rightarrow \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-150}}$

- soft max $[0, 1]$

- ReLU $[0, \infty]$

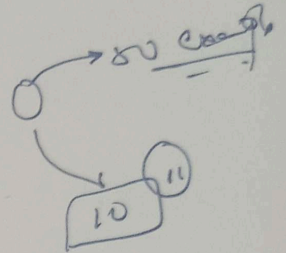
- tanh

- Soft plus



0

0



Early Stopping ($\text{Validation loss} < \text{Training loss}$)

Drop out :

Regularization

Data Augmentation

RNN

LSTM $\rightarrow$ i/p $\rightarrow$ $\rightarrow$ out/p $\rightarrow$ $\rightarrow$ send

$$\text{ReLU} = \max(0, x)$$

$$f(x) = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases}$$

Cat Sat on the mat. (text)



["cat", "sat", "on", "the", "mat"]



[1, 2, 3, 4, 5]



[0.255022, ...]

[0.25...][...][...]



[0.25...][...][...][...]



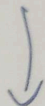
Attend

$$\text{Attention}(Q, K, V) = \text{Softmax}$$

$$\left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

Set of probability.

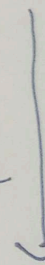
① Tokenisation



② Token id



③ Embedding



④ positional Encoding

Process in the transformer

$$\text{Softmax}(x_i) = \frac{e^{x_i}}{\sum_{j=1}^n e^{x_j}}$$

Temp = (x)

$$e^1 = 2.718 \quad /$$

$$e^2 = 7.389 \quad /$$

$$e^3 = 20.086 \quad /$$

top K

top B

Training an LLMs:

① Data

② Compute / Infra

③ Architecture

④ Evals

⑤ Objective - Response

Text $\rightarrow$ Tokens $\rightarrow$ Embedding $\rightarrow$ positional embedding $\rightarrow$ encoder

formula

[Attention]

Softmax $\leftarrow$ Linear - Decoder

① Tokenisation

② Embedding

③ Position Embedding

④ $\left\{ \begin{array}{l} \text{attention scores} \\ \text{feed forward} \end{array} \right\}$ encoder

⑤ Decoder - masked
multi-head
states.

Finetuning

→ Supervised FT

→ parameter efficient fine
low Rank Adaptation

[low Rank
matrix] Adapter

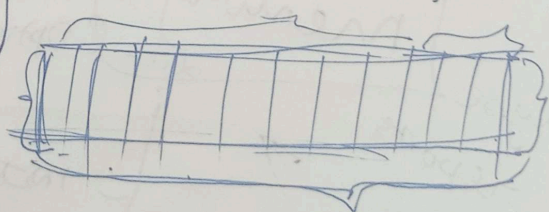
finetuning →

LoRA

QLoRA

Quantized

IE - medical
.org
needs



RAG $\rightarrow$ Similarity Search (happens inside Search)
 Vector Search
 $\downarrow$
 GRAPH RAG
 $\downarrow$
 Hybrid RAG.

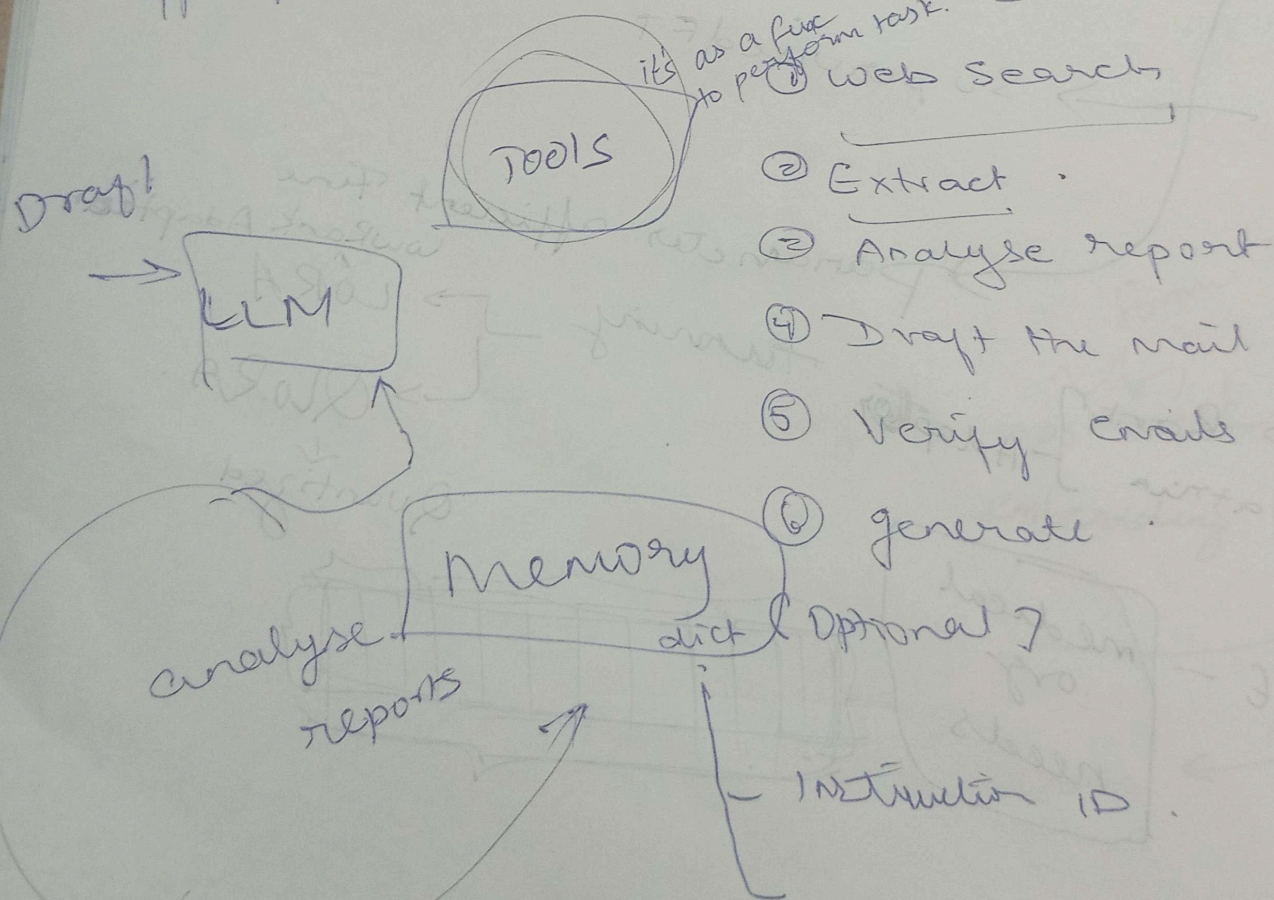
ATTENTION MECHANISM

Q

K

V

GPT Architecture $\rightarrow$ Decoder only



$$a_0 = 12.42$$

$$a_1 = -5.51$$

$$a_2 = 0.76$$



classification under

Logistic Regression in ML:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$z = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$$

Bayes Theorem:

It predicts the class of a new data pt by calculating probability. It is called naive

dimension of

$m \times n$

Query (Q) = what you are looking for

Key (k) =

Value (v) =

